

PAVAN ALAKUNTA

AI & Machine Learning Engineer

pavanalakunta58@gmail.com | +44 7721318451 | linkedin.com/in/pavan-alakunta | github.com/Pavan09-Is-Here | Luton, UK

PROFESSIONAL SUMMARY

AI and Machine Learning graduate with hands-on experience building AI products, LLM/RAG systems, and full-stack applications end to end. Comfortable working across the AI stack, from data quality and model evaluation to backend APIs, databases, and modern frontend interfaces. Strong mathematical foundations in linear algebra, probability, and optimisation. Seeking an AI Engineer role to deliver scalable, high-impact AI solutions within a UK technology organisation.

TECHNICAL SKILLS

Programming Languages:	Python, C, R, MATLAB, SQL, TypeScript, JavaScript
ML Frameworks:	PyTorch, TensorFlow, Keras, Scikit-learn, Hugging Face Transformers
Computer Vision:	OpenCV, YOLOv8, YOLOv4
Natural Language Processing:	BERT, GPT fine-tuning, NLTK, Sentence Transformers, RAG Pipelines
AI Product Engineering:	FastAPI, SQLAlchemy, Next.js, React, Supabase, OpenAI API
Data Engineering:	Pandas, NumPy, Matplotlib, Seaborn, Plotly, PostgreSQL
Development Tools:	Linux (Ubuntu), Jupyter Notebook, VS Code, PyCharm, Git
Mathematics:	Linear Algebra, Multivariate Calculus, Bayesian Inference, Probability Theory

WORK EXPERIENCE

AI Engineer

Centific, Hyderabad, India | 2025

- Worked on NLP, AI training, and generative AI data quality tasks across LLM datasets.
- Reviewed dataset quality, task consistency, and output accuracy across annotation pipelines.
- Designed and applied prompt engineering strategies that improved model response quality to above 98% accuracy.
- Evaluated and refined AI model responses, supporting RLHF workflows used to fine-tune LLMs.
- Supported structured workflows for improving AI model responses across NLP and generative AI tasks.

Freelance AI Trainer & Subject Matter Expert

Chegg / Outlier, Remote | 2023 - 2024

- Provided expert-level solutions for complex technical and mathematical academic queries.
- Collaborated on AI training and data refinement projects for LLMs, ensuring 100% accuracy against strict deadlines.

AWS AI & ML Virtual Intern

AICTE - AWS Academy, India | 2023

- Implemented cloud-based ML workflows covering data preprocessing, algorithm selection, and model deployment.
- Focused on building reproducible pipelines and foundational deployment practices on AWS.

KEY PROJECTS

LearnForge AI

AI-Powered Learning Platform | 2026

- Built an AI education platform focused on real understanding, structured learning responses, RAG-based retrieval, chat memory, and PDF generation.
- Designed the product idea, user flow, and AI learning experience end to end.
- Built backend APIs using Python, FastAPI, and the OpenAI API, with Supabase for conversation and user data.
- Developed the frontend using Next.js, TypeScript, and Tailwind CSS.

CareerPilot AI

Career Automation Platform | 2026

- Built a personal career automation platform around one hard rule: the system never invents candidate facts.
- Implemented a verified facts vault and truthful, ATS-friendly CV generation traceable to verified records.
- Built compliance-gated job discovery, explainable eligibility matching, and safety gates for CAPTCHA, MFA, and irreversible submissions.
- Stack: FastAPI, SQLAlchemy 2, Next.js 16, React 19, TypeScript, PostgreSQL/Supabase.

Deepfake Video Detection using Deep Learning

Final Year Project | 2025

- Technologies: Python, TensorFlow, PyTorch, OpenCV, CNN/LSTM, FaceForensics++ dataset.
- Implemented a hybrid CNN-LSTM architecture to detect spatial and temporal inconsistencies in deepfake videos.
- Trained on FaceForensics++ and Celeb-DF datasets using transfer learning with an EfficientNet backbone.
- Deployed as a Flask web application with real-time video processing capability.

Crop Protection from Animal Intrusion using Machine Learning

Academic Project | 2024

- Technologies: Python, TensorFlow/Keras, OpenCV, Raspberry Pi, IoT Sensors.
- Developed a real-time animal intrusion detection system using CNNs to prevent crop damage.
- Trained a CNN on custom datasets of wild boar, deer, and other animal imagery, integrated with IoT sensors and Raspberry Pi for edge deployment.

EDUCATION

MSc Artificial Intelligence and Robotics

University of Hertfordshire, Hatfield, UK | Jan 2026 - Present

- Key Modules: Artificial Life with Robotics, Neural Networks and Machine Learning, Theory and Practice of Artificial Intelligence.
- Activities: University AI and Robotics Society.

B.Tech Artificial Intelligence and Machine Learning

Nalla Malla Reddy Engineering College, Hyderabad, India | Oct 2021 - May 2025

- Classification: First Class Honours | GPA: 7.68 / 10.00.
- Final Year Project: Deepfake Video Detection using Deep Learning.
- Core Modules: Data Structures and Algorithms, OOP, Linear Algebra, Probability and Statistics, Computer Networks, AI, Machine Learning, Deep Learning, NLP, Software Engineering.

CERTIFICATIONS

- AWS AI/ML Virtual Internship - AICTE | Completed 2023

ADDITIONAL INFORMATION

Languages:	English (Fluent), Telugu (Native), Hindi (Proficient)
Right to Work:	UK Student Visa holder - eligible for UK Graduate Route Visa providing 2 years post-study work authorisation upon graduation.
Interests:	Large Language Models and Generative AI, AI Ethics and Model Evaluation, Emerging AI Technologies, Cooking, Basketball.